Sahaj Bhadja

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Summary

Software Engineer and USC MSCS graduate building **performance-focused C++ and Python systems**, interactive developer tools, simulations, automated tests, and reliable cross-platform applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Data Structures, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Web Technologies, Computer Networks, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Investigated application and database bottlenecks for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through **performance analysis**, indexed SQL queries, validation checks, code reviews, and debugging.
- Reworked asynchronous Python services with FastAPI workers, explicit job states, retries, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **software reliability** in production.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent Python services for secure file ingestion and multimodal processing, handling 1K+ files per batch with fault isolation, provider fallbacks, and **automated testing** across production workflows.
- Built React and Flask features supporting 150+ instructors and 10K+ daily requests, translating requirements into maintainable interfaces and REST APIs while reducing **workflow turnaround** 60% during peak usage.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving trajectory planning 18% while processing 100K+ logs through reproducible **Python simulation and modeling** pipelines for controlled experiments.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **performance evaluation**, debugging, failure analysis, and documented iterative research reviews.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform integration modules using object-oriented design patterns, improving API response times 30%, reducing technical debt 20%, and strengthening **application compatibility** across Windows, Linux, and macOS deployments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code reviews**, defect resolution, Git collaboration, and dependable releases across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight MERN microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **interactive React interfaces**, REST APIs, WebSockets, and robust data validation across distributed locations.
- Developed TypeScript workflows for real-time records and third-party integrations, while Jenkins CI/CD, Docker staging, and **automated regression testing** safely reduced deployment time 50% across controlled production releases.

Skills

Languages: C++, Python, C, Java, JavaScript, TypeScript, SQL, Bash, HTML5, CSS3
Systems & Performance: RISC-V, MIPS, Computer Architecture, Microprocessors, Simulation, Performance Tuning, Profiling, Benchmarking, Multithreading, Concurrency, Data Structures, Algorithms, Linux, Unix, Memory Management
Application Development: React, Node.js, Flask, FastAPI, REST APIs, WebSockets, PostgreSQL, SQL Server, MongoDB, Data Visualization, Responsive Design, OOP, Design Patterns
Engineering: Git, GitHub, Docker, CI/CD, Jenkins, GitHub Actions, Unit Testing, Integration Testing, Regression Testing, PyTest, Mockito, JUnit, Code Review, Debugging, Agile, SDLC, Technical Documentation, AI-Assisted Development

Projects

RISC-V Performance Studio | *C++, Python, React, WebAssembly, Linux Perf*

- Designed an interactive developer tool that profiles RISC-V workloads, visualizes instruction hotspots and cache behavior, compares optimization runs, and applies **performance-analysis workflows** through a React interface and C++ engine.

Microarchitecture Tuning Lab | *C++, RISC-V, Python, Cache Simulation, GoogleTest*

- Built a configurable RISC-V simulator modeling pipelines, branch prediction, and cache policies, with Python experiment automation, **benchmark-driven tuning**, regression tests, trace diagnostics, and reproducible performance comparison results.

CPU-Scheduling Simulator Live | *C++, React, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue algorithms in C++ with **interactive performance visualization**, validating 1K+ process combinations and reducing average turnaround time up to 25%.